

ECO387C - Fall 2024

Problem Set 8 Solutions

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You may work in groups, but please acknowledge any contributions from others and turn in separate assignments.

1 Endowment economy with a cooler

There are two periods: period 0 and period 1, and one good. There is only one type of agent, a household. Assume that the total mass of agents is $N = 1$. In period 0, each agent seeks to maximize the expected lifetime utility.

$$\max \mathbb{E}_0 \sum_{t=0}^1 \beta^t u(c_t)$$

In addition to the risk-free asset (a) with an interest rate of $1 + r$, the agent has another means of saving: cooler (x). The agent can save consumption goods in the cooler now and take them out for consumption in the next period. Thus, the agent is subject to the following budget constraints:

$$\begin{aligned} c_0 + x + a &= y_0 \\ c_1 &= y_1 + x + (1 + r)a \\ x &\geq 0 \end{aligned}$$

1. Write down the agent's maximization problem as a Lagrangian. Find the Euler equations.

The Lagrangian is

$$\begin{aligned} \mathcal{L} &= u(c_0) + \beta u(c_1) + \lambda_0 [y_0 - a - x - c_0] \\ &\quad + \lambda_1 [y_1 + a(1 + r) + x - c_1] \\ &\quad + \theta x \end{aligned}$$

The FONCs are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_0} = 0 &\Rightarrow u'(c_0) - \lambda_0 = 0 \\ \frac{\partial \mathcal{L}}{\partial c_1} = 0 &\Rightarrow \beta u'(c_1) - \lambda_1 = 0 \\ \frac{\partial \mathcal{L}}{\partial x} = 0 &\Rightarrow -\lambda_0 + \lambda_1 + \theta = 0 \\ \frac{\partial \mathcal{L}}{\partial a} = 0 &\Rightarrow -\lambda_0 + (1 + r)\lambda_1 = 0 \end{aligned}$$

The Euler equation for the risk-free asset is

$$u'(c_0) = \beta(1+r)u'(c_1)$$

The Euler equation for the cooler is

$$u'(c_0) = \beta u'(c_1) + \theta$$

Now consider two cases:

- (a) If $x = 0$ ($\theta > 0$): This reduces to the standard endowment economy. But note that the first-order conditions of x and a are consistent only when $r > 0$ and $\theta = ru'(c_1) > 0$. Otherwise, we would have a negative multiplier. The Euler equations for both cooler and the risk-free asset are

$$u'(c_0) = \beta(1+r)u'(c_1)$$

- (b) If $x > 0$ ($\theta = 0$): In this case, the first-order conditions of x and a are consistent only when $r = 0$. The Euler equations for both cooler and the risk-free asset are

$$u'(c_0) = \beta u'(c_1)$$

2. Suppose $\beta = 1$ and $u(c) = \log(c)$. Find the equilibrium consumption and interest rates in this economy considering the following two endowment profiles:

- $(y_0, y_1) = (4, 3)$
- $(y_0, y_1) = (3, 4)$

We need to consider two possible cases:

- (a) If $x = 0$: The market clearing conditions are

$$c_0 = y_0, \quad c_1 = y_1, \quad a = 0$$

The interest rate consistent with the allocation is

$$(1+r) = \frac{c_1}{c_0} = \frac{y_1}{y_0}$$

- If $(y_0, y_1) = (4, 3)$, $(1+r) = \frac{3}{4}$: This means $r = -1/4 < 0$. This would be the equilibrium interest rate without the cooler. However, with the cooler, this cannot constitute an equilibrium since it violates the individual optimality condition by making the multiplier θ negative.
- If $(y_0, y_1) = (3, 4)$, $(1+r) = \frac{4}{3}$: This means $r = 1/3 > 0$. This would be the equilibrium interest rate with $x^* = 0$.

- (b) If $x > 0$: The market clearing conditions are

$$c_0 + x = y_0, \quad c_1 = y_1 + x, \quad a = 0$$

The equilibrium interest rate consistent with this allocation should be $r = 0$. The equilibrium saving in the cooler is then

- If $(y_0, y_1) = (4, 3)$

$$1 = \frac{c_1}{c_0} = \frac{3+x}{4-x} \Leftrightarrow x^* = \frac{1}{2} > 0$$

- If $(y_0, y_1) = (3, 4)$

$$1 = \frac{c_1}{c_0} = \frac{4+x}{3-x} \Leftrightarrow x^* = -\frac{1}{2} < 0$$

Since x needs to be non-negative, this cannot constitute an equilibrium.

Thus, the equilibrium in this economy for each endowment profile is

- If $(y_0, y_1) = (4, 3)$:

$$(c_0^*, c_1^*, x^*, a^*, 1+r^*) = (3.5, 3.5, 0.5, 0, 1)$$

- If $(y_0, y_1) = (3, 4)$:

$$(c_0^*, c_1^*, x^*, a^*, 1+r^*) = (3, 4, 0, 0, \frac{4}{3})$$

3. Repeat the previous question with $u(c) = -\frac{1}{2}(c - \bar{c})^2$ where $\bar{c} >> 0$ is some large number representing a satiation point.

Following the same procedure:

- If $(y_0, y_1) = (4, 3)$:

$$(c_0^*, c_1^*, x^*, a^*, 1+r^*) = (3.5, 3.5, 0.5, 0, 1)$$

- If $(y_0, y_1) = (3, 4)$:

$$(c_0^*, c_1^*, x^*, a^*, 1+r^*) = (3, 4, 0, 0, \frac{3-\bar{c}}{4-\bar{c}})$$

2 Endowment economy without a cooler

Consider the same optimization problem of the agent but suppose now that there is no cooler.

Part 1: Without uncertainty Consider the endowment profile $(y_0, y_1) = (4, 3)$

1. Write down the agent's maximization problem as a Lagrangian. Find the Euler equations.

The Euler equation is

$$u'(c_0) = \beta(1+r)u'(c_1)$$

2. Suppose $\beta = 1$ and $u(c) = \log(c)$. Find the equilibrium consumption and interest rates $(1 + r_{\log,1}^*)$ in this economy.

Due to the market clearing condition, the equilibrium interest rate is

$$(1 + r_{\log,1}^*) = \frac{u'(c_0)}{u'(c_1)} = \frac{y_1}{y_0} = \frac{3}{4}$$

3. Suppose $\beta = 1$ and $u(c) = -\frac{1}{2}(c - \bar{c})^2$ where $\bar{c} > 0$ is some large number representing a satiation point. Find the equilibrium consumption and interest rates $(1 + r_{\text{quad},1}^*)$ in this economy.

Due to the market clearing condition, the equilibrium interest rate is

$$(1 + r_{\text{quad},1}^*) = \frac{u'(c_0)}{u'(c_1)} = \frac{-(y_0 - \bar{c})}{-(y_1 - \bar{c})} = \frac{4 - \bar{c}}{3 - \bar{c}}$$

Part 2: With uncertainty Suppose now that the period 1 endowment is uncertain while the period 0 endowment remains the same at 4. Specifically, the period 1 endowment will be either high $y_1^h = 5$ or low $y_1^l = 1$. The probability of getting a high endowment is $1/2$.

1. Write down the agent's maximization problem as a Lagrangian. Find the Euler equation.

The Euler equation is

$$u'(c_0) = \beta(1+r)\mathbb{E}_0[u'(c_1)]$$

2. Suppose $\beta = 1$ and $u(c) = \log(c)$. Find the equilibrium consumption and interest rates $(1 + r_{\log,2}^*)$ in this economy. How does $(1 + r_{\log,2}^*)$ compare with $(1 + r_{\log,1}^*)$?

$$(1 + r_{\log,2}^*) = \frac{u'(c_0)}{\mathbb{E}_0[u'(c_1)]} = \frac{\frac{1}{y_0}}{\frac{1}{2}(\frac{1}{y_1^h} + \frac{1}{y_1^l})} = \frac{\frac{1}{4}}{\frac{1}{2}(\frac{1}{5} + \frac{1}{1})} = \frac{5}{12}$$

We can see that $(1 + r_{\log,1}^*) > (1 + r_{\log,2}^*)$. Introducing uncertainty lowers the equilibrium interest rate because of the precautionary saving motive. This is because the marginal utility of the log utility function is convex (prudence).

3. Suppose $\beta = 1$ and $u(c) = -\frac{1}{2}(c - \bar{c})^2$ where $\bar{c} > 0$ is some large number representing a satiation point. Find the equilibrium consumption and interest rates $(1 + r_{\text{quad},2}^*)$ in this economy. How does $(1 + r_{\text{quad},2}^*)$ compare with $(1 + r_{\text{quad},1}^*)$?

$$(1 + r_{\text{quad},2}^*) = \frac{u'(c_0)}{\mathbb{E}_0[u'(c_1)]} = \frac{-(4 - \bar{c})}{\frac{1}{2}(- (5 - \bar{c}) - (1 - \bar{c}))} = \frac{4 - \bar{c}}{3 - \bar{c}}$$

We can see that $(1 + r_{\text{quad},1}^*) = (1 + r_{\text{quad},2}^*)$. Introducing uncertainty does not affect the equilibrium interest rate because the quadratic utility function lacks prudence.